16/April/26 * EXPERIMENT - 13 *

* **AIM** :- Determining the probability of a given problem using negative binomial distribution.

* **EXPERIMENT** :- If a shooter aims at a target, obtain the probability that the 5th fire is the second hit given that probability of hitting the target at any trial is 0.20.

* **THEORY** :- A random variable 'X' is said to follow negative binomial distribution if its PMF is given by ;

$$\Rightarrow \boxed{P[X=x] = \binom{x+K-1}{K-1} p^K q^x}$$

where x = specific number of failures
K = fixed number of successes
$x+K$ = total trials

* **RESULT** :- The probability that the 5th hit fire is the 2nd hit is 0.08192

CALCULATION

→ Probability of hit $= p = 0.20$

→ Probability of miss $= q = 1 - p = 0.8$

$k = 2$ (second hit)

$n = 5 - 2 = 3$ (failures)

Using negative binomial distribution,

$$\Rightarrow P[X=3] = \binom{3+2-1}{2-1} p^2 q^3$$

$$= {}^4C_1 (0.2)^2 (0.8)^3$$

$$= \frac{4!}{3!} \times 0.04 \times 0.512$$

$$\therefore \boxed{P[X=3] = 0.08192}$$